

Coverage-Oriented Patrol Routing for Rhino Anti-Poaching in Etosha National Park: A Team-Orienteering Model on a Real Road Network

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Abstract

Background: Rhino poaching in Namibia intensified through the mid-2020s, and Etosha National Park absorbed a large share of the losses. Once deterrents such as dehorning and arrests are in place, the remaining operational lever is patrol efficiency: given limited vehicles, fuel, and ranger-hours, where should patrols drive? **Methods:** Patrol planning was formulated as a multi-depot team orienteering problem routed on the real OpenStreetMap road network. Each cell of a 5 km grid carried a reward equal to a modelled poaching-risk score times a resource-based habitat-integrity index, both taken from an antecedent spatial model of the park (Model-1). Patrols were time-budgeted loops from ranger camps, solved heuristically with Google OR-Tools guided local search, benchmarked against four baselines, and tested for sensitivity to the candidate set, budget, and risk-map error. The travel-time model was checked against Model-1's own travel-time field. **Results:** Five eight-hour patrols traversed cells covering 26% of the mapped risk-reward and 70–80% of the highest-priority cells, exceeding greedy, risk-only, habitat-only, and random baselines and remaining essentially unchanged under 15% risk-map noise. A marginal-value test found that a second patrol adds most at the western Okaukuejo and Dolomite stations. **Conclusion:** The routes concentrate patrol time where risk and habitat coincide and tolerate modest input error. This is a modelling prototype: the risk and habitat fields are modelled, the travel model is checked only against another model, and the heuristic solution carries no optimality guarantee. Validation against observed ranger movement, operational constraints, and randomisation are required before deployment.

Keywords: anti-poaching; conservation optimization; vehicle routing with profits; mixed-integer programming; guided local search; wildlife protection; geographic information systems

1 Introduction

Illegal poaching drives species loss and degrades protected ecosystems, and rhinoceros populations are among the most heavily targeted; the black rhino (*Diceros bicornis*) is listed as Critically Endangered,¹ and the high illicit value of rhino horn sustains demand for poaching.² Namibia experienced escalating rhino poaching through the mid-2020s, and Etosha National Park, one of the country's largest reserves, absorbed a disproportionate share of the losses.³

Interventions such as dehorning, which sharply reduces poaching pressure,⁴ and sustained anti-poaching patrols, whose effectiveness is well documented,⁵ reduce losses, but once those interventions are in place the remaining operational lever is patrol efficiency. Rangers operate with a fixed number of vehicles, a limited fuel allowance, and a bounded working day, so the practical decision each day is where a patrol should drive.

A large body of work frames anti-poaching patrol planning as an adversarial optimisation problem in which a

defender commits to a patrol strategy that an attacker then best-responds to.^{6,7} The Protection Assistant for Wildlife Security (PAWS) and its predecessors model the poacher as a strategic adversary in a Stackelberg security game and compute patrols that are hard to predict,^{8,9} and the green security games literature extends this to attackers with limited surveillance and to behaviour learned from real poaching records.^{10,11} These systems have been deployed through conservation platforms¹² and are informed by ranger-collected data on the spatial patterns of illegal activity.¹³ Less attention has been paid to the complementary question of how to route a fixed set of patrols on the actual road network so as to maximise the risk they cover, treating the risk surface as given. This paper addresses that question for Etosha.

The central modelling decision is a deliberate inversion of conventional route planning. Ordinary routing minimises travel effort or steers around hazard; the model developed here instead maximises the hazard that patrols cover. Each cell of a discretised map carries a reward equal to the poaching risk it represents multiplied by a resource-based proxy for whether a rhino is likely to be present, and the patrols are chosen to collect as much of this reward as possible within a time budget. This is the structure of the team orienteering problem (TOP), in which a fixed number of tours gathers the largest possible prize subject to a per-tour length limit.¹⁴ The objective of this study is to build, exercise, and stress-test such a model on real Etosha geography, to benchmark it against simpler strategies, and to state honestly what would be required before it could inform real patrols.

2 Methods

2.1 Study Area and Data

Etosha National Park covers approximately 22,270 km² and is dominated by the Etosha Pan, a large salt pan that is poor rhino habitat and slow to traverse. The geographic layers—roads, boundary, gates, and waterholes—are real and georeferenced; the two decision-driving fields, poaching risk and habitat integrity, are modelled (Section 2.2), not measured. The analytical core was a grid of $N = 805$ cells at 5 km resolution. Each cell i carried a modelled poaching-risk score $\rho_i \in [0, 1]$ and a habitat-integrity index $h_i \in [0, 1]$ (both defined in Section 2.2), together with land cover, the normalised difference vegetation index (NDVI),¹⁵ distances to the boundary, road, waterhole, and nearest ranger base, and a base-to-cell travel time τ_i^{ref} . The park boundary and the Etosha Pan were supplied as real polygons. The patrol infrastructure comprised five ranger camps (Okaukuejo, Halali, Namutoni, Dolomite, and Onkoshi), four entry gates, and fifty-five waterholes described by type and seasonality. The road network was a set of 419 real segments from OpenStreetMap,¹⁶ classified as primary, secondary, tertiary,

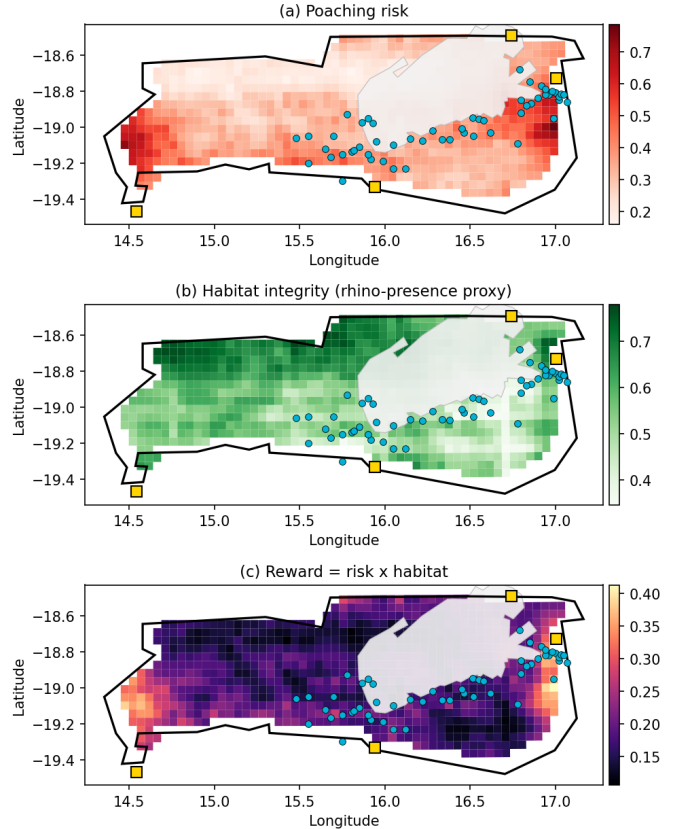


Figure 1: Real input surfaces on the 805-cell grid, drawn at the park’s true geographic aspect ratio: (a) poaching risk ρ , (b) habitat integrity h used as a rhino-presence proxy, and (c) the patrol reward $r = \rho h$. Black outline: park boundary; grey: Etosha Pan; cyan discs: waterholes; yellow squares: gates.

or unclassified. Empirically ρ_i had a median of 0.33 and a maximum of 0.79, and h_i had a median of 0.56; the derived reward defined below had a median of 0.18 and a maximum of 0.41, so reward was spread thinly with a light upper tail. Figure 1 shows the three surfaces.

2.2 The Model-1 Grid

The two decision-driving fields, poaching risk ρ and habitat integrity h , are not produced by this study. They are outputs of an antecedent, multi-module spatial model of Etosha (referred to here as *Model-1*) that discretises the park into the 805 cells of a 5 km grid and is built from ESA WorldCover land cover, MODIS NDVI, CHIRPS rainfall, NASA FIRMS active-fire data, GBIF species occurrences, OpenStreetMap,¹⁶ and the World Database on Protected Areas. This paper treats Model-1’s per-cell fields as fixed inputs and contributes the routing model built on top of them.

The poaching-risk score is a bounded human-threat proxy that is highest where illegal entry and ambush are

easiest,

$$\rho_i = 0.35 e^{-d_i^{\text{gate}}/20} + 0.25 e^{-d_i^{\text{edge}}/25} + 0.20 e^{-d_i^{\text{WH}}/8} + 0.20 \text{RoadDens}_i, \quad (1)$$

where d_i^{gate} , d_i^{edge} , and d_i^{WH} are the distances in kilometres from cell i to the nearest gate, park boundary, and waterhole, and RoadDens_i is normalised road density. The habitat-integrity index is $h_i = 1 - D_i^{\text{hab}}$, where $D_i^{\text{hab}} \in [0, 1]$ is Model-1's chronic habitat-disturbance score. D_i^{hab} aggregates resource sub-indices, including forage adequacy ($0.5 \text{NDVI} + 0.3 \text{grassland} + 0.2(1 - \text{pan})$), cover ($0.5 \text{woodland} + 0.3 \text{shrub} + 0.2(1 - \text{bare})$), and water reliability, each normalised to $[0, 1]$; the full derivation is documented in Model-1 and reproduced in the supplement. Because ρ and h are modelled quantities rather than field measurements, they are treated throughout as informed priors, and the sensitivity and robustness analyses probe how the routing results depend on them.

2.3 Reward Model

The reward assigned to a cell must be large only where a patrol both deters a likely threat and protects a likely animal. The patrol reward was therefore defined as the product of the poaching-risk score and a rhino-presence proxy,

$$r_i = \rho_i h_i, \quad \rho_i, h_i \in [0, 1], \quad (2)$$

where ρ_i is the modelled poaching risk in cell i and h_i is the habitat-integrity index. Because the product is high only when both factors are high, the reward concentrates on cells that combine threat with good habitat. The index h_i is a resource-based proxy for habitat quality, not a measurement of rhino presence. Habitat-suitability studies for black rhino, such as the assessment of the Ngorongoro Conservation Area by Borges et al.,¹⁷ show that remote-sensing proxies capture broad suitability but must be validated against field observations of shelter and breeding sites; that study concerns Tanzania and cannot validate an Etosha model directly, so h_i should be read as an informed prior rather than ground truth. Two coverage notions were used throughout. A cell is a *selected stop* if the solver routes a patrol to it as a visit, and a cell is *traversed* if a patrol passes through it between stops. Because a patrol provides deterrent presence wherever it drives, the traversed set is the more faithful measure of coverage, and results are reported for both.

2.4 Movement Graph and Travel-Time Cost

Patrol effort was measured in time rather than distance, because a working-day limit is the operational constraint a warden faces. Routing was performed on a graph built

from the real road network. Every vertex of every OpenStreetMap road segment became a node, and shared vertices acted as junctions, so a route follows real road geometry. A road edge between consecutive vertices was weighted by its length divided by a class-dependent speed, with 70, 60, 45, and 35 km/h for primary, secondary, tertiary, and unclassified roads; these class-dependent values follow the road-class and land-cover speed assumptions used in global accessibility mapping¹⁸ and are consistent with the posted limit on Etosha's main roads. Two kinds of off-road links were then added so that patrols were not confined to roads. Each cell centroid was joined to its nearest road vertex, and adjacent cell centroids within 7.6 km were joined to each other; both used an off-road speed

$$s_i^{\text{off}} = \begin{cases} 15 \text{ km/h,} & \text{salt-pan cell,} \\ 32 \text{ km/h,} & \text{other terrain,} \end{cases} \quad (3)$$

so that crossing the pan is expensive and cross-country travel elsewhere is possible but slower than driving a road. Because Etosha's road data include tracks that cross the pan, the same 15 km/h pan speed is also applied to road edges whose endpoints fall inside the mapped pan polygon, so a patrol gains no speed advantage from a nominal road across the pan. The travel time between two cells a and b was the shortest-path time

$$T(a, b) = \min_{P \in \mathcal{P}_{ab}} \sum_{(i,j) \in P} \frac{d_{ij}}{v_{ij}}, \quad (4)$$

where d_{ij} is the great-circle length of edge (i, j) , v_{ij} is its speed, and $\mathcal{P}_{ab}$ is the set of paths from a to b ; it was computed with Dijkstra's algorithm¹⁹ over the combined road-and-off-road graph, which for this study had about 23,900 nodes. These shortest-path times formed the arc costs of the routing problem, so a patrol prefers roads and skirts the pan without any additional rule.

2.5 Team Orienteering Formulation

Let M be the set of patrols, one per active ranger station, with patrol $m \in M$ based at depot cell b_m , and let C be the set of candidate reward cells. A binary variable x_{ij}^m equals one when patrol m traverses arc (i, j) , a binary variable y_i^m equals one when patrol m visits cell i , and continuous variables u_i^m carry visit order. The model is

$$\max \sum_m \sum_{i \in C} r_i y_i^m - \lambda \sum_m \sum_{(i,j)} T(i, j) x_{ij}^m \quad (5)$$

$$\text{s.t. } \sum_m y_i^m \leq 1, \quad \forall i \in C, \quad (6)$$

$$\sum_j x_{b_m j}^m = \sum_j x_{j b_m}^m = 1, \quad \forall m, \quad (7)$$

$$\sum_j x_{ji}^m = \sum_j x_{ij}^m = y_i^m, \quad \forall m, i, \quad (8)$$

$$\sum_{(i,j)} T(i, j) x_{ij}^m \leq T_{\max}, \quad \forall m, \quad (9)$$

$$u_i^m - u_j^m + |C| x_{ij}^m \leq |C| - 1, \quad \forall m, i, j, \quad (10)$$

with $x_{ij}^m, y_i^m \in \{0, 1\}$ and $u_i^m \in \mathbb{R}$. The objective in Eq. (5) is an explicit weighted objective, not a pure budget-constrained maximisation. The first term is the collected reward and the second is a small penalty on travel time with weight λ . In the implementation, reward was scaled by $R = 10^6$ and travel time was expressed in seconds, so the effective weight is $\lambda = 1/R = 10^{-6}$ reward-units per second, equivalently 3.6×10^{-3} reward-units per hour. Because $\lambda T_{\max} = 2.9 \times 10^{-2}$ reward-units is below the smallest candidate reward (0.05), travel can never make it worthwhile to drop a positive-reward cell; it only shortens routes among options that collect the same reward. Constraint (6) lets each cell count for at most one patrol; Eq. (7) forces each patrol to leave and return to its base; Eq. (8) couples the arc and visit variables; Eq. (9) enforces the per-patrol time budget; and Eq. (10) is the Miller–Tucker–Zemlin subtour-elimination condition,²⁰ which keeps each patrol a single connected loop.

2.6 Solution Method and Experiments

The TOP contains the single-vehicle orienteering problem²¹ as a special case and is NP-hard,^{14,22} so exact solution is impractical for a candidate set of order 10^2 cells with five vehicles under interactive time limits; solution approaches are reviewed in dedicated surveys.²³ Common metaheuristics for the problem include tabu search,²⁴ ant colony optimisation,²⁵ and other neighbourhood- and population-based methods.²⁶ The formulation also belongs to the broader family of routing problems with profits, alongside the prize-collecting travelling salesman problem²⁷ and travelling salesman problems with profits.²⁸ The model was solved with the Google OR-Tools routing engine²⁹ (version 9.15.6755), a general constraint-based vehicle-routing solver,³⁰ using the guided-local-search metaheuristic under a fixed per-solve time limit. Optional visits were represented as disjunctions in which skipping a candidate incurs a penalty equal to its reward, which realises the reward term of Eq. (5); the arc cost realises the small travel term; and the budget of Eq. (9) was imposed on a routing dimension. All-pairs shortest-path times among candidate and depot cells were computed once with a multi-source Dijkstra routine. After the solver returned, each pair of consecutive stops was expanded into its full shortest path so that the drawn route follows the road graph and the traversed cells can be scored.

Unless noted, the reported study used five depots, a candidate set of the $n = 130$ highest-reward cells, and a budget of $T_{\max} = 8$ h (a working day). Four baselines were evaluated under the same five patrols and budget, all scored by traversed reward against the true reward of Eq. (2). A *risk-only* and a *habitat-only* run used the same solver but selected candidates and set disjunction penalties from ρ_i or h_i alone. A *greedy nearest-hotspot* heuristic defined the hot set as candidate cells with reward at or above the median and, from each base, repeatedly

appended the time-nearest unvisited hot cell whose insertion kept the loop within budget. A *random feasible loops* baseline built loops by sampling random candidates and accepting a candidate when the loop stayed within budget, averaged over 25 random seeds. The marginal value of capacity was measured by adding one more eight-hour patrol at a single base at a time and recording the extra traversed reward. Sensitivity was assessed by varying the candidate cutoff ($n \in \{80, 130, 200\}$) and the time budget ($T_{\max} \in \{4, 6, 8, 10, 12\}$ h), and robustness to risk-map error by adding independent Gaussian noise with a standard deviation of 15% to the reward and re-solving eight times.

2.7 Travel-Time Comparison

The off-road speed in Eq. (3) is the main free parameter of the cost model. It was set so that the graph’s base-to-cell shortest times $\min_m T(b_m, i)$ resemble Model-1’s own travel-time field τ_i^{ref} across all 805 cells. Because τ_i^{ref} is derived from the same road data, this is an internal consistency check rather than an independent calibration; the residual is reported in Section 3.5.

3 Results

3.1 Main Plan

With five patrols, a top-130 candidate set, and an eight-hour budget, the plan traversed cells worth 38.7 of the 149.8 reward available, or 26%, while the reward at selected stops alone was 26.1 (17%); the difference is the deterrent value of the ground a patrol covers between stops. Coverage concentrated on the worst cells: the traversed routes covered 80%, 72%, and 70% of the top 25, 50, and 100 highest-reward cells. Figure 2 shows the loops on the real road network, and Table 1 reports each patrol. The routes skirt the salt pan, except for the Onkoshi patrol, whose camp sits on the pan’s northern shore so that some pan travel is unavoidable. Namutoni is the most productive station per hour; Dolomite and Onkoshi also perform well.

3.2 Baselines

Table 2 compares the team-orienteering solver against the four baselines. The solver captured the most traversed reward, although its margin over risk-only selection is small; both clearly exceed a greedy nearest-hotspot heuristic, habitat-only selection, and random feasible loops. That risk-only is competitive is expected, because poaching risk is the larger and more variable of the two reward factors here; the value of the product reward is clearest in the sizeable gap over habitat-only selection. Because the solver is a time-limited guided local search on an NP-hard problem, these are strong feasible solutions, not certified optima; the

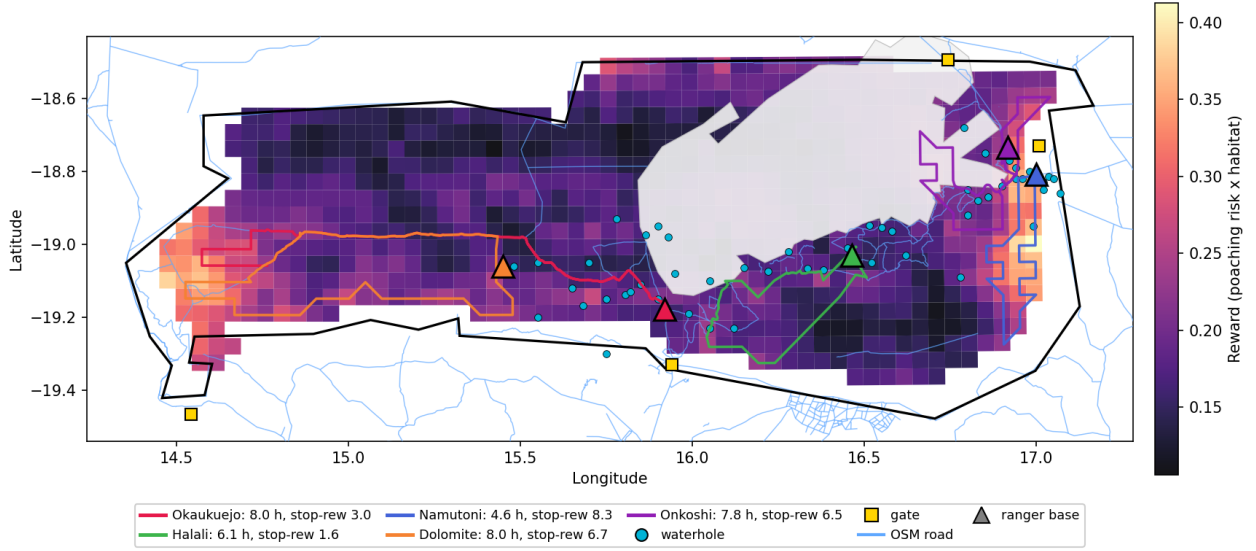


Figure 2: Planned patrol routes on the real OpenStreetMap road network with an eight-hour budget per patrol. Coloured loops start and end at ranger bases (triangles) and follow real roads (blue), skirting the salt pan (grey). Cell colour is the reward $r = \rho h$; cyan discs are waterholes and yellow squares are gates.

Table 1: Per-patrol performance under an eight-hour budget. Stop reward counts each patrol’s selected cells once; traversed cells count all cells driven through. Per-patrol traversed counts overlap between patrols, so they exceed the 165 unique cells covered; the stop rewards sum exactly to the total because no cell is selected by two patrols (Eq. 6).

Ranger base	Time (h)	Stop rew.	Trav. cells
Namutoni	4.6	8.3	28
Dolomite	8.0	6.7	49
Onkoshi	7.8	6.5	35
Okaukuejo	8.0	3.0	44
Halali	6.1	1.6	28
Total	34.5	26.1	165 (unique)

comparison establishes that it beats the baselines, not that it is optimal.

3.3 Marginal Value of an Added Patrol

Table 3 adds a second eight-hour patrol at one base at a time and measures the extra traversed reward. A second patrol adds positive reward at every base, from +3.2 to +5.6, so extra capacity helps wherever it is placed. The gains are largest at the western Okaukuejo and Dolomite stations and smallest at Onkoshi, whose first patrol is already confined by the pan to the north-eastern shore. Because the range across stations is narrow, this is a soft preference for western reinforcement rather than a strong ranking.

Table 2: Traversed reward captured by the team-orienting solver and four baselines (five patrols, eight hours). Percentages are of the 149.8 available.

Method	Trav. reward	Share
Team orienteering (risk $\times$ hab.)	38.7	26%
Risk-only selection	37.7	25%
Greedy nearest-hotspot	31.2	21%
Habitat-only selection	27.1	18%
Random feasible loops	23.0 ± 2.4	15%

Table 3: Extra traversed reward from adding one more eight-hour patrol at each base, relative to the five-patrol plan.

Base of added patrol	Δ traversed reward
Okaukuejo	+5.6
Dolomite	+5.2
Halali	+4.3
Namutoni	+4.1
Onkoshi	+3.2

3.4 Sensitivity and Robustness

The candidate cutoff mattered mainly below saturation: a top-80 set captured 30.1 reward, top-130 captured 38.5, and top-200 captured 40.7, so most of the achievable reward is reached by 130, with a few per cent more available from a larger set. The time budget produced the concave curve in Figure 3, rising from 14% of available reward at four hours to 26% at eight hours and 34% at ten hours, after which twelve hours (36%) added little because the

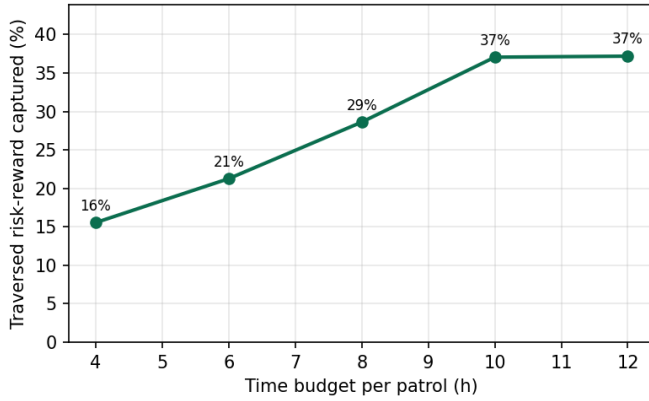


Figure 3: Traversed risk-reward captured as a function of the per-patrol time budget, with five patrols and a top-130 candidate set. Returns are concave and saturate near ten hours.

candidate set saturates. Robustness to error in the risk map was strong: adding independent Gaussian noise with a standard deviation of 15% to the reward and re-solving eight times left the traversed reward essentially unchanged at 38.5 ± 1.3 , so the plan is not sensitive to modest error in the risk layer.

3.5 Travel-Time Comparison

The graph and Model-1 travel-time fields agree in central tendency—medians 1.59 versus 1.36 h, with a small positive bias (+0.11 h)—but differ cell by cell. The mean absolute error (MAE) is 0.68 h, the root-mean-square error (RMSE) is 0.91 h, and the Pearson correlation is 0.50, so the travel model explains about a quarter (25%) of the per-cell variance; the interquartile ranges are [0.99, 2.35] and [0.81, 2.35] h. This is a model-against-model comparison, not a validation against observed ranger movement, and the modest per-cell correlation remains a limitation of the only empirical anchor available.

4 Discussion

On real geography and a real road network, a team orienteering formulation concentrated patrol time on the boundary corridors where risk and habitat coincide, exceeded four sensible baselines, and was robust to modest error in the risk layer. The marginal-value test shows that a second patrol helps at every station, most in the west (Okaukuejo and Dolomite) and least at Onkoshi, though the differences between stations are small. These findings are internally consistent: stress testing reproduced every headline number and verified that all patrols respect the budget, return to base, and never double-count reward, and the per-patrol stop rewards sum exactly to the reported total.

The most important methodological caveat concerns

the objective. The solver maximises the reward of *selected stops* (the y_i variables), while success is judged mainly by *traversed* reward. The two are related because the drawn route is expanded to its full path and scored cell by cell, but they are not identical, and a route can pass through a high-reward cell without the objective crediting it. In exploratory runs, treating every cell as a reward-bearing decision variable so that a visited cell equals a traversed cell aligned the solved and reported quantities and gave comparable total coverage; formalising that coverage variable and optimising it directly is the natural next step and is left for future work.

Other gaps separate this prototype from an operational tool. The risk and habitat fields are model estimates and would be sharpened by live incident data and by a habitat model fitted and validated with Etosha field observations rather than borrowed from another park. The travel model is calibrated only against another model; validation against held-out ranger GPS or travel logs, reporting error, bias, and quantiles on a test split, is the necessary next step and was not possible here because such data were unavailable. The eight-hour budget is treated as driving time and omits dwell and search time at a stop, rest breaks, seasonal road closures, gate opening hours, vehicle range, and ranger safety constraints; each can be represented as an extra term in the time budget or as a side constraint. Finally, a single optimal plan is predictable, which an adversary can exploit; because the reward surface is fixed, the deterministic model can serve as the defender’s action space inside a Stackelberg security game,^{6,8,10,11} and sampling several high-reward loops rather than always driving the best one yields a randomised schedule that preserves expected coverage while denying prediction. Such randomisation should precede any real deployment.

5 Conclusion

This study formulated rhino anti-poaching patrol planning for Etosha National Park as a multi-depot team orienteering problem that routes on the real OpenStreetMap road network with explicit off-road links. The reward couples a modelled poaching-risk score to a resource-based habitat proxy, both from an antecedent model (Model-1), and the cost is a travel time whose residual against Model-1’s own travel-time field is reported. Solved heuristically with guided local search, the routes concentrate scarce ranger-hours on the corridors where risk and habitat coincide, traverse cells covering about 70–80% of the highest-priority cells with five eight-hour patrols, beat risk-only, habitat-only, greedy, and random baselines, and are essentially unchanged under 15% risk-map noise. A marginal-value test finds that a second patrol helps most in the west (Okaukuejo and Dolomite), with only modest differences between stations. The results should be read as a modelling prototype rather than a deployable plan; the next

steps are to optimise traversed coverage directly, to validate the travel model against observed ranger movement, to add operational time constraints, and to introduce a randomisation layer, all of which fit the same formulation.

Abbreviations

GIS: geographic information system; GLS: guided local search; GPS: global positioning system; IQR: interquartile range; MAE: mean absolute error; MTZ: Miller–Tucker–Zemlin; NDVI: normalised difference vegetation index; OP: orienteering problem; OSM: OpenStreetMap; PAWS: Protection Assistant for Wildlife Security; RMSE: root-mean-square error; SMART: Spatial Monitoring and Reporting Tool; TOP: team orienteering problem.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics Statement

Ethics approval was not required because the study used only publicly available data and did not involve human participants, private identifiable data, or animal subjects.

Data Availability Statement

All source code, input data files, and the scripts that produce every table and figure are provided in Supplementary File 1. The road, boundary, and waterhole layers are from OpenStreetMap and are available under the Open Database License; the curated analytical grid is derived from public remote-sensing and geographic sources described in the Methods.

AI Use Statement

The authors used Anthropic’s Claude and OpenAI’s ChatGPT for software development, data processing, figure generation, literature look-up, and drafting. All AI-generated content and computational results were extensively reviewed, verified, and edited by the authors, who take full responsibility for the accuracy and integrity of the

manuscript. Verification included re-running all analysis scripts to reproduce every reported number, auditing the arithmetic in each table, fixing several of the mistakes that AI had made, and cross-checking every reference against its DOI or source URL.

Authors’ Contributions

T.M. (first author) conceived and led the study and carried out the majority of the work, including the conceptualization, methodology, software implementation, formal analysis, visualization, and writing of the manuscript. R.S., I.K., and N.G. contributed to background research, data checking, and review of the manuscript. Authorship is not equal: the first author’s contribution substantially exceeded that of the other authors. All authors read and approved the final manuscript.

Acknowledgements

The authors thank the OpenStreetMap contributors for the geographic data used in this study.

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Supplementary Material

Supplementary File 1 (the Supplementary Material folder): all Python source code (`real_model.py`, `osm_routing.py`, `osm_solver.py`, `osm_viz.py`, `experiments.py`), the input data (grid, boundary, salt pan, roads, ranger bases, gates, waterholes), the machine-readable results (`experiments.json`), and a README with exact commands to regenerate every table and figure in this paper.